Depropanizer Column Ver 1 · OTS

DC3-V1-IN-INL-001

Instrument list and I/O count

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
Document number	DC3-V1-IN-INL-001
Title	Instrument list and I/O count
Discipline	Instrumentation
Originator / DRAWN	Apinan Soottitantawat
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Source	simulator model (design snapshot, alarm list, trip constants, HMB runs); tabs/einst/work/inl.json

Revision record

REV	DATE	DESCRIPTION	BY	CHK	APP
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Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

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1 · Purpose, scope and basis

This document is the instrument list (instrument index) of the C-201 depropanizer unit: every instrument of the P&ID DC3-V1-PID-001 with its service, type and measuring principle, range and units, the normal value at the design point of the simulator model, the set point, the alarm limits, the trip setting or relief set pressure, the I/O type (AI, AO, DI, DO), the system it is wired to (DCS, SIS, local, field or mechanical), its fail position and its P&ID line or equipment. It holds 109 rows plus 4 recommended additions from the safety-layers review (marked “rec.”, not counted in the base I/O). It also gives the I/O count with the spare allowance, the control-valve list with the C_v required in the three HMB cases and the SIS / BPCS separation check. Trip effects are in the cause and effect matrix DC3-V1-IN-CE-001 and the SIF requirements in DC3-V1-IN-SRS-001; they are not repeated here.

Sources

- Tags: P&ID bubbles (BUB), DCS indicators (IND), controllers (sim.C), SIS and auto-start functions of the model; first draft from epc/epc.json refined by tabs/einst/work/mk_inl.js (changes in section 7).
- Alarm limits: the simulator alarm list ALM. Trip settings: model constants K.Lsll, K.Lshh, K.fsll, K.Psv; PDSL-230 from the model logic (150 kPa, 3 s).
- Normal values: the design snapshot of the model (warm start 3 000 s, noise off). Control-valve cases: the design snapshot and two runs of copies of the model (maximum feed, turndown; 14 400 s) as in DC3-V1-PE-HMB-001.
- Fail positions: P&ID note 5, confirmed on a copy of the model with instrument air failed (see the Instrumentation tab).

Abbreviations

Code	Meaning	Code	Meaning
AI / AO	analog input / output (4–20 mA)	DCS	distributed control system (BPCS)
DI / DO	discrete input / output	SIS	safety instrumented system (separate logic solver)
FC / FO	fail closed / fail open	rec.	recommended addition (not in the base scope)
L / H	low / high alarm	Trip / set	trip setting and delay, or relief set pressure

2 · Instrument list

Tag	Service	Type / principle	Range	Units	Normal (design)	SP	Alarm L	Alarm H	Trip / set	I/O	System	Fail	P&ID line / equipment	Loop	Note
FE-201	Feed to C-201	Flow element (orifice plate)	—	m³/h	—	—	—	—	—	—	Field	—	4"-P-202	FIC-201	
FT-201	Feed to C-201	Transmitter, orifice + DP transmitter	0–100	m³/h	60.0	—	—	—	—	AI	DCS	—	4"-P-202	FIC-201	
FIC-201	Feed to C-201	DCS controller (AUTO at design; Kc 0.3, Ti 6 s)	0–100	m³/h	60.0	60.0	45.0	70.0	—	—	DCS	—	4"-P-202	FIC-201	
FV-201	Feed to C-201	Control valve, diaphragm actuator	0–100	%	40.0	—	—	—	—	AO	DCS	Fail closed	4"-P-202	FIC-201	
TT-202	Feed preheat (E-201 outlet)	Transmitter, RTD in thermowell	0–150	°C	75.0	—	—	—	—	AI	DCS	—	6"-P-203 / 6"-LS-301	TIC-202	
TIC-202	Feed preheat (E-201 outlet)	DCS controller (AUTO at design; Kc 0.8, Ti 60 s)	0–150	°C	75.0	75.0	68.0	85.0	—	—	DCS	—	6"-P-203 / 6"-LS-301	TIC-202	
TV-202	Feed preheat (E-201 outlet)	Control valve, diaphragm actuator	0–100	%	50.0	—	—	—	—	AO	DCS	Fail closed	6"-P-203 / 6"-LS-301	TIC-202	
FE-204	Reflux to tray 27	Flow element (orifice plate)	—	m³/h	—	—	—	—	—	—	Field	—	4"-P-209	FIC-204	
FT-204	Reflux to tray 27	Transmitter, orifice + DP transmitter	0–100	m³/h	49.0	—	—	—	—	AI	DCS	—	4"-P-209	FIC-204	
FIC-204	Reflux to tray 27	DCS controller (AUTO at design; Kc 0.3, Ti 6 s)	0–100	m³/h	49.0	49.0	35.0	65.0	—	—	DCS	—	4"-P-209	FIC-204	
FV-204	Reflux to tray 27	Control valve, diaphragm actuator	0–100	%	50.0	—	—	—	—	AO	DCS	Fail open	4"-P-209	FIC-204	
TT-227	Top tray 27 temperature (cascade to FIC-204, normally MAN)	Transmitter, RTD in thermowell	0–100	°C	45.0	—	—	—	—	AI	DCS	—	C-201 tray 27	TIC-227	
TIC-227	Top tray 27 temperature (cascade to FIC-204, normally MAN)	DCS controller (MAN at design; Kc 2, Ti 600 s)	0–100	°C	45.0	45.0	—	50.0	—	—	DCS	—	C-201 tray 27	TIC-227	
TT-208	Tray 8 temperature → reboiler steam	Transmitter, RTD in thermowell	0–150	°C	76.9	—	—	—	—	AI	DCS	—	C-201 tray 8 / 6"-LS-302	TIC-208	
TIC-208	Tray 8 temperature → reboiler steam	DCS controller (AUTO at design; Kc 3, Ti 900 s)	0–150	°C	76.9	76.9	72.0	82.0	—	—	DCS	—	C-201 tray 8 / 6"-LS-302	TIC-208	
TV-208	Tray 8 temperature → reboiler steam	Control valve, diaphragm actuator	0–100	%	50.0	—	—	—	—	AO	DCS	Fail closed	C-201 tray 8 / 6"-LS-302	TIC-208	
LT-202	C-201 sump level → bottoms	Transmitter, DP level transmitter	0–100	%	50.0	—	—	—	—	AI	DCS	—	C-201 sump / 4"-P-210	LIC-202	
LIC-202	C-201 sump level → bottoms	DCS controller (AUTO at design; Kc 1.2, Ti 600 s)	0–100	%	50.0	50.0	25.0	75.0	—	—	DCS	—	C-201 sump / 4"-P-210	LIC-202	
LV-202	C-201 sump level → bottoms	Control valve, diaphragm actuator	0–100	%	40.0	—	—	—	—	AO	DCS	Fail closed	C-201 sump / 4"-P-210	LIC-202	
LT-203	V-202 level → propane product	Transmitter, DP level transmitter	0–100	%	50.0	—	—	—	—	AI	DCS	—	V-202 / 3"-P-213	LIC-203	
LIC-203	V-202 level → propane product	DCS controller (AUTO at design; Kc 1.2, Ti 600 s)	0–100	%	50.0	50.0	25.0	75.0	—	—	DCS	—	V-202 / 3"-P-213	LIC-203	
LV-203	V-202 level → propane product	Control valve, diaphragm actuator	0–100	%	40.0	—	—	—	—	AO	DCS	Fail closed	V-202 / 3"-P-213	LIC-203	
PT-210	Column top pressure → vent to fuel gas	Transmitter, pressure transmitter	0–2.5	MPa g	1.70	—	—	—	—	AI	DCS	—	8"-P-204 / 1"-P-206	PIC-210	
PIC-210	Column top pressure → vent to fuel gas	DCS controller (AUTO at design; Kc 6, Ti 400 s)	0–2.5	MPa g	1.70	1.70	1.60	1.76	—	—	DCS	—	8"-P-204 / 1"-P-206	PIC-210	
PV-210	Column top pressure → vent to fuel gas	Control valve, diaphragm actuator	0–100	%	15.0	—	—	—	—	AO	DCS	Fail open	8"-P-204 / 1"-P-206	PIC-210	
LT-201	V-201 level (rev B indication, rev C cascade to FIC-201)	Transmitter, DP level transmitter	0–100	%	50.0	—	—	—	—	AI	DCS	—	V-201	LIC-201	rev A: local only (LG-201); rev B/C on the DCS
LI-201	V-201 level indication (rev B and rev C)	DCS indication	0–100	%	50.0	—	20.0	80.0	—	—	DCS	—	V-201	LI-201	rev B: LAL/LAH-201 on the DCS
LIC-201	V-201 level (rev B indication, rev C cascade to FIC-201)	DCS controller (MAN at design; Kc 1, Ti 900 s)	0–100	%	50.0	50.0	—	—	—	—	DCS	—	V-201	LIC-201	rev C only
TT-201	Feed temperature from V-201	Temperature transmitter (RTD)	−20–40	°C	−1.00	—	—	—	—	AI	DCS	—	6"-P-201	TI-201	
TI-201	Feed temperature from V-201	DCS indication	−20–40	°C	−1.00	—	—	—	—	—	DCS	—	6"-P-201	TI-201	
FT-206	LP steam to E-201	Flow transmitter (orifice + DP)	0–10	t/h	5.82	—	—	—	—	AI	DCS	—	6"-LS-301	FI-206	
FI-206	LP steam to E-201	DCS indication	0–10	t/h	5.82	—	—	—	—	—	DCS	—	6"-LS-301	FI-206	
PT-207	LP steam header pressure	Pressure transmitter	0–600	kPa g	450	—	—	—	—	AI	DCS	—	6"-LS-302	PI-207	
PI-207	LP steam header pressure	DCS indication	0–600	kPa g	450	—	350	—	—	—	DCS	—	6"-LS-302	PI-207	
TT-241	Tray 1 temperature	Temperature transmitter (RTD)	0–130	°C	107	—	—	—	—	AI	DCS	—	C-201 tray 1	TI-241	
TI-241	Tray 1 temperature	DCS indication	0–130	°C	107	—	—	—	—	—	DCS	—	C-201 tray 1	TI-241	
TT-242	Tray 4 temperature	Temperature transmitter (RTD)	0–130	°C	98.1	—	—	—	—	AI	DCS	—	C-201 tray 4	TI-242	
TI-242	Tray 4 temperature	DCS indication	0–130	°C	98.1	—	—	—	—	—	DCS	—	C-201 tray 4	TI-242	
TT-243	Tray 12 temperature	Temperature transmitter (RTD)	0–130	°C	67.8	—	—	—	—	AI	DCS	—	C-201 tray 12	TI-243	
TI-243	Tray 12 temperature	DCS indication	0–130	°C	67.8	—	—	—	—	—	DCS	—	C-201 tray 12	TI-243	
TT-245	Tray 18 (feed tray) temperature	Temperature transmitter (RTD)	0–130	°C	65.8	—	—	—	—	AI	DCS	—	C-201 tray 18	TI-245	
TI-245	Tray 18 (feed tray) temperature	DCS indication	0–130	°C	65.8	—	—	—	—	—	DCS	—	C-201 tray 18	TI-245	
TT-246	Tray 22 temperature	Temperature transmitter (RTD)	0–130	°C	52.1	—	—	—	—	AI	DCS	—	C-201 tray 22	TI-246	
TI-246	Tray 22 temperature	DCS indication	0–130	°C	52.1	—	—	—	—	—	DCS	—	C-201 tray 22	TI-246	
TT-209	Sump temperature	Temperature transmitter (RTD)	0–130	°C	109	—	—	—	—	AI	DCS	—	C-201 sump	TI-209	
TI-209	Sump temperature	DCS indication	0–130	°C	109	—	—	—	—	—	DCS	—	C-201 sump	TI-209	
PT-211	Column top pressure, independent	Pressure transmitter	0–2.5	MPa g	1.70	—	—	—	—	AI	DCS	—	8"-P-204	PI-211	
PI-211	Column top pressure, independent	DCS indication	0–2.5	MPa g	1.70	—	—	1.78	—	—	DCS	—	8"-P-204	PI-211	
PDT-212	Column ΔP, sump to top	Differential pressure transmitter	0–120	kPa	17.9	—	—	—	—	AI	DCS	—	C-201	PDI-212	
PDI-212	Column ΔP, sump to top	DCS indication	0–120	kPa	17.9	—	—	30.0	—	—	DCS	—	C-201	PDI-212	
FT-205	LP steam to E-203	Flow transmitter (orifice + DP)	0–8	t/h	3.79	—	—	—	—	AI	DCS	—	6"-LS-302	FI-205	
FI-205	LP steam to E-203	DCS indication	0–8	t/h	3.79	—	—	—	—	—	DCS	—	6"-LS-302	FI-205	
PT-205	E-203 steam chest pressure	Pressure transmitter	−80–500	kPa g	254	—	—	—	—	AI	DCS	—	E-203 shell	PI-205	
PI-205	E-203 steam chest pressure	DCS indication	−80–500	kPa g	254	—	—	—	—	—	DCS	—	E-203 shell	PI-205	
FT-202	Bottoms to the debutanizer	Flow transmitter (orifice + DP)	0–80	m³/h	27.8	—	—	—	—	AI	DCS	—	4"-P-210	FI-202	
FI-202	Bottoms to the debutanizer	DCS indication	0–80	m³/h	27.8	—	—	—	—	—	DCS	—	4"-P-210	FI-202	
AT-220	C3 and lighter in the bottoms	On-line analyzer (GC), sample system	0–20	mol %	1.00	—	—	—	—	AI	DCS	—	4"-P-210	AI-220	
AI-220	C3 and lighter in the bottoms	DCS indication	0–20	mol %	1.00	—	—	2.00	—	—	DCS	—	4"-P-210	AI-220	

Tag	Service	Type / principle	Range	Units	Normal (design)	SP	Alarm L	Alarm H	Trip / set	I/O	System	Fail	P&ID line / equipment	Loop	Note
FT-203	Propane product	Flow transmitter (orifice + DP)	0–80	m³/h	32.1	—	—	—	—	AI	DCS	—	3"-P-213	FI-203	
FI-203	Propane product	DCS indication	0–80	m³/h	32.1	—	—	—	—	—	DCS	—	3"-P-213	FI-203	
AT-219	C4 and heavier in the propane	On-line analyzer (GC), sample system	0–20	mol %	1.50	—	—	—	—	AI	DCS	—	3"-P-213	AI-219	
AI-219	C4 and heavier in the propane	DCS indication	0–20	mol %	1.50	—	—	2.50	—	—	DCS	—	3"-P-213	AI-219	
TT-213	Condensate / reflux temperature	Temperature transmitter (RTD)	−10–80	°C	39.4	—	—	—	—	AI	DCS	—	6"-P-205	TI-213	
TI-213	Condensate / reflux temperature	DCS indication	−10–80	°C	39.4	—	—	—	—	—	DCS	—	6"-P-205	TI-213	
FT-216	Cooling water to E-202	Flow transmitter (orifice + DP)	0–400	m³/h	330	—	—	—	—	AI	DCS	—	10"-CW-401	FI-216	
FI-216	Cooling water to E-202	DCS indication	0–400	m³/h	330	—	250	—	—	—	DCS	—	10"-CW-401	FI-216	
TT-214	CW supply temperature	Temperature transmitter (RTD)	0–50	°C	15.0	—	—	—	—	AI	DCS	—	10"-CW-401	TI-214	
TI-214	CW supply temperature	DCS indication	0–50	°C	15.0	—	—	—	—	—	DCS	—	10"-CW-401	TI-214	
TT-215	CW return temperature	Temperature transmitter (RTD)	0–60	°C	25.3	—	—	—	—	AI	DCS	—	10"-CW-402	TI-215	
TI-215	CW return temperature	DCS indication	0–60	°C	25.3	—	—	—	—	—	DCS	—	10"-CW-402	TI-215	
FT-217	Vent gas to fuel gas	Flow transmitter (orifice + DP)	0–800	Nm³/h	72.5	—	—	—	—	AI	DCS	—	1"-P-206	FI-217	
FI-217	Vent gas to fuel gas	DCS indication	0–800	Nm³/h	72.5	—	—	—	—	—	DCS	—	1"-P-206	FI-217	
PDI-230	P-202 differential pressure	DCS indication	0–800	kPa	551	—	—	—	—	—	DCS	—	4"-P-208	PDI-230	
FE-202	Bottoms to the debutanizer	Flow element (orifice plate)	—	—	—	—	—	—	—	—	Field	—	4"-P-210	FI-202	
FE-203	Propane product	Flow element (orifice plate)	—	—	—	—	—	—	—	—	Field	—	3"-P-213	FI-203	
FE-205	LP steam to E-203	Flow element (orifice plate)	—	—	—	—	—	—	—	—	Field	—	6"-LS-302	FI-205	
FE-206	LP steam to E-201	Flow element (orifice plate)	—	—	—	—	—	—	—	—	Field	—	6"-LS-301	FI-206	
FE-216	Cooling water to E-202	Flow element (orifice plate)	—	—	—	—	—	—	—	—	Field	—	10"-CW-401	FI-216	
FE-217	Vent gas to fuel gas	Flow element (orifice plate)	—	—	—	—	—	—	—	—	Field	—	1"-P-206	FI-217	
LS-202	C-201 sump level, trip sensor (separate nozzle)	Level transmitter to the SIS (drawn LS; 4–20 mA, trip in the logic solver)	0–100	%	50.0	—	—	—	< 5 %	AI	SIS	—	C-201 sump	LSLL-202	
LSLL-202	Sump low-low → close XV-209 and XV-202 (heat off)	SIS trip function	—	%	50.0	—	—	—	< 5 % for 5 s	—	SIS	—	C-201 sump	LSLL-202	armed only with the SIS in RUN
LS-203	V-202 level, trip sensor (separate nozzle)	Level transmitter to the SIS (drawn LS; 4–20 mA, trip in the logic solver)	0–100	%	50.0	—	—	—	> 92 %	AI	SIS	—	V-202	LSHH-203	
LSHH-203	V-202 high-high → close XV-201, XV-209, XV-202	SIS trip function	—	%	50.0	—	—	—	> 92 % for 5 s	—	SIS	—	V-202	LSHH-203	
HS-ESD	Emergency shutdown push-button (feed and heat off)	Hand switch (console)	—	—	—	—	—	—	manual	DI	SIS	—	Control room	ESD	
HS-RST	SIS reset push-button (console): re-opens XV-201 after a trip	Hand switch (console, hard-wired)	—	—	—	—	—	—	—	DI	SIS	—	Control room	ESD	reset blocked while LSHH-203 is active
FSLL-201	Low feed flow → auto-start P-201B (not a trip)	Switch function on FT-201	0–100	m³/h	60.0	—	—	—	< 15 m³/h for 5 s	—	SIS	—	4"-P-202	FSLL-201	only with FV-201 > 20 % and XV-201 open
PDT-230	P-202 differential pressure	Differential pressure transmitter	0–800	kPa	551	—	—	—	—	AI	SIS	—	4"-P-208	PDSL-230	
PDSL-230	Low P-202 ΔP → auto-start P-202B (not a trip)	Switch function on PDT-230	0–800	kPa	551	—	—	—	< 150 kPa for 3 s	—	SIS	—	4"-P-208	PDSL-230	
XV-201	Feed shut-off	On/off valve, spring-return actuator + solenoid	open/closed	—	—	—	—	—	closes on trip	DO	SIS	Fail closed	4"-P-202	SIS	
ZSC-201	Feed shut-off — closed limit switch	Limit switch	—	—	—	—	—	—	—	DI	SIS	—	4"-P-202	SIS	
XV-202	Preheater steam shut-off	On/off valve, spring-return actuator + solenoid	open/closed	—	—	—	—	—	closes on trip	DO	SIS	Fail closed	6"-LS-301	SIS	
ZSC-202	Preheater steam shut-off — closed limit switch	Limit switch	—	—	—	—	—	—	—	DI	SIS	—	6"-LS-301	SIS	
XV-209	Reboiler steam shut-off	On/off valve, spring-return actuator + solenoid	open/closed	—	—	—	—	—	closes on trip	DO	SIS	Fail closed	6"-LS-302	SIS	
ZSC-209	Reboiler steam shut-off — closed limit switch	Limit switch	—	—	—	—	—	—	—	DI	SIS	—	6"-LS-302	SIS	
HY-201A	Feed pump P-201A start/stop command to the MCC	Interposing relay output (MCC starter)	—	—	—	—	—	—	—	DO	DCS	—	P-201A	P-201A	
YI-201A	Feed pump P-201A run status	Run feedback	—	—	—	—	—	—	—	DI	DCS	—	P-201A	P-201A	
HY-201B	Feed pump (spare) P-201B start command (trip-and-throttle valve solenoid)	Solenoid output (turbine start)	—	—	—	—	—	—	—	DO	DCS	—	P-201B	P-201B	
YI-201B	Feed pump (spare) P-201B run status	Run feedback	—	—	—	—	—	—	—	DI	DCS	—	P-201B	P-201B	
HY-202A	Reflux pump P-202A start/stop command to the MCC	Interposing relay output (MCC starter)	—	—	—	—	—	—	—	DO	DCS	—	P-202A	P-202A	
YI-202A	Reflux pump P-202A run status	Run feedback	—	—	—	—	—	—	—	DI	DCS	—	P-202A	P-202A	
HY-202B	Reflux pump (spare) P-202B start/stop command to the MCC	Interposing relay output (MCC starter)	—	—	—	—	—	—	—	DO	DCS	—	P-202B	P-202B	
YI-202B	Reflux pump (spare) P-202B run status	Run feedback	—	—	—	—	—	—	—	DI	DCS	—	P-202B	P-202B	
XA-201A	Feed pump P-201A motor tripped (MCC protection relay)	Fault contact (MCC)	—	—	—	—	—	—	—	DI	DCS	—	P-201A	P-201A	alarm on the DCS
XA-202A	Reflux pump P-202A motor tripped (MCC protection relay)	Fault contact (MCC)	—	—	—	—	—	—	—	DI	DCS	—	P-202A	P-202A	alarm on the DCS
XA-202B	Reflux pump P-202B motor tripped (MCC protection relay)	Fault contact (MCC)	—	—	—	—	—	—	—	DI	DCS	—	P-202B	P-202B	alarm on the DCS
HS-201B	Auto-start enable of P-201B on FSLL-201	Hand switch (console)	—	—	—	—	—	—	—	—	DCS	—	P-201B	FSLL-201	
HS-202B	Auto-start enable of P-202B on PDSL-230	Hand switch (console)	—	—	—	—	—	—	—	—	DCS	—	P-202B	PDSL-230	
LG-201	V-201 level gauge glass	Level gauge (local)	0–100	%	—	—	—	—	—	—	Local	—	V-201	—	rev A: the only V-201 level indication
PSV-201	C-201 overpressure to flare	Pressure safety valve 4"×6", orifice L on the P&ID rev D (1841 mm²); M (2323 mm²) required by DC3-V1-PE-RLB-001, see ER-01	—	MPa g	—	1.80	—	—	set 1.80 MPa g	—	Mechanical	—	8"-P-204	—	
PT-221 rec.	Column top pressure, SIS transmitter	Pressure transmitter (SIS)	0–2.5	MPa g	1.70	—	—	—	—	AI	SIS	—	8"-P-204	PSHH-221	recommended addition (Safety layers tab)

Tag	Service	Type / principle	Range	Units	Normal (design)	SP	Alarm L	Alarm H	Trip / set	I/O	System	Fail	P&ID line / equipment	Loop	Note
PSHH-221 rec.	Column pressure high-high → close XV-209 and XV-202 (heat off)	SIS trip function	—	MPa g	1.70	—	—	—	> 1.77 MPa g for 3 s (assumed, Safety layers tab)	—	SIS	—	C-201 overhead	PSHH-221	recommended addition (Safety layers tab)
XV-222 rec.	Bottoms shut-off on LSLL-202	On/off valve, spring-return actuator + solenoid	open/closed	—	—	—	—	—	—	DO	SIS	Fail closed	4"-P-210	LSLL-202	recommended addition (Safety layers tab)
ZSC-222 rec.	Bottoms shut-off — closed limit switch	Limit switch	—	—	—	—	—	—	—	DI	SIS	—	4"-P-210	LSLL-202	recommended addition (Safety layers tab)

Items of the first draft moved to the valve list (piping items, not instruments): ST-201 (E-201 condensate), ST-202 (E-203 condensate), HV-202 (6"-LS-301), HV-209 (6"-LS-302), HV-210 (1"-P-206), HV-216 (10"-CW-401).

3 · I/O count and spare allowance

Hard-wired signals only (controllers and indications are DCS software). Installed spare 20 % of the used I/O, rounded up per type (assumed); I/O cards of 16 channels (DCS) and 8 channels (SIS) (assumed). Base scope: **60 I/O**, 75 installed with spares, 9 cards. With the recommended additions: 63 I/O.

System	I/O type	Used	Spare (20 %)	Installed	I/O cards	Channels	Free channels
DCS	AI	31	7	38	3	48	35 %
DCS	AO	7	2	9	1	16	56 %
DCS	DI	7	2	9	1	16	56 %
DCS	DO	4	1	5	1	16	75 %
DCS total		**49**	**12**	**61**	**6**	**96**	**49 %**
SIS	AI	3	1	4	1	8	62 %
SIS	AO	0	0	0	0	0	0 %
SIS	DI	5	1	6	1	8	38 %
SIS	DO	3	1	4	1	8	62 %
SIS total		**11**	**3**	**14**	**3**	**24**	**54 %**
Unit total		**60**	**15**	**75**	**9**	**120**	**50 %**

4 · Control valve list

C_v by IEC 60534-2-1 (turbulent flow, F_p = 1). Liquid: $K_v = (Q / N_1)\cdot\sqrt{(G / \Delta P)}$ with ΔP capped at the choked limit $F_L^2\cdot(P_1 - F_F\cdot P_v)$, $F_F = 0.96 - 0.28\cdot\sqrt{(P_v/P_c)}$. Gas and steam: $K_v = W / (N_6\cdot Y\cdot\sqrt{(x\cdot P_1\cdot\rho_1)})$, $Y = 1 - x / (3\cdot F_\gamma\cdot x_T)$, x capped at $F_\gamma\cdot x_T$. $C_v = 1.156\cdot K_v$. Service conditions from the model relations of each valve (flow, upstream and downstream pressure, opening). Characteristic: linear when the valve ΔP is nearly constant over the cases ($\Delta P_{min}/\Delta P_{max} \geq 0.5$), otherwise equal percentage (R = 50). Rated C_v: smallest standard trim ≤ 90 % open at the largest required C_v. Window: 10–90 % (linear), 20–90 % (equal %).

Tag	Service	Line	Fail	Fluid	Regime D / M / T	Cv required D / M / T	Char.	Rated Cv	% open (selected) D / M / T	Model valve position D / M / T	Model Cv at 100 %	Cv max : min	Window
FV-201	Feed to C-201 (after P-201A/B)	4"-P-202	FC	liquid	normal / normal / normal	24.9 / 37.3 / 13.1	equal %	63	76 / 87 / 60	40 / 60 / 21	62.1 / 62.1 / 62.1	2.8 : 1	✓
TV-202	LP steam to E-201	6"-LS-301	FC	steam	sub-critical / sub-critical / choked (critical)	99.4 / 130.5 / 63.1	equal %	200	82 / 89 / 70	50 / 71 / 32	198.9 / 182.8 / 196.3	2.1 : 1	✓
FV-204	Reflux to tray 27	4"-P-209	FO	liquid	normal / normal / normal	19.7 / 24.7 / 11.9	linear	32	62 / 77 / 37	50 / 63 / 30	39.4 / 39.3 / 39.7	2.1 : 1	✓
TV-208	LP steam to E-203	6"-LS-302	FC	steam	sub-critical / sub-critical / sub-critical	71.1 / 90.3 / 44.5	linear	125	57 / 72 / 36	50 / 67 / 29	142.2 / 134.9 / 155.4	2.0 : 1	✓
LV-202	Bottoms to the debutanizer	4"-P-210	FC	liquid	choked, flashing / choked, flashing / choked, flashing	11.4 / 13.3 / 7.6	linear	16	71 / 83 / 48	40 / 47 / 27	28.5 / 28.5 / 28.5	1.7 : 1	✓
LV-203	Propane product to storage	3"-P-213	FC	liquid	choked, flashing / choked, flashing / choked, flashing	8.6 / 10.1 / 5.3	linear	12.5	69 / 81 / 42	40 / 46 / 26	21.4 / 22.1 / 20.5	1.9 : 1	✓
PV-210	Vent gas to fuel gas	1"-P-206	FO	gas	choked (critical) / choked (critical) / choked (critical)	0.33 / 2.3 / 0.10	linear	4	8 / 57 / 2	15 / 91 / 6	2.2 / 2.5 / 1.7	22.8 : 1	✗

Worked example — FV-201 at the design point

- Q = 60.00 m³/h (FI-201), valve 40.0 % open; ΔP from the model relation $F = K_{v1}\cdot x\cdot\sqrt{\Delta P} = (60.00 / (7.259\cdot 0.400))^2$ = 427 kPa.
- Liquid at −1 °C: ρ 548.5 kg/m³ (G 0.5485), P_v 413.6 kPa a; P_1 = pump discharge 2,363 kPa g. Not choked (ΔP_{choked} ≈ 1 700 kPa).
- $K_v = (60 / 0.1)\cdot\sqrt{(0.5485 / 427)}$ = 21.5 m³/h; C_v = 24.86. Equal-percentage trim C_v 63: 76 % open at design, 87 % at maximum feed, 60 % at turndown. The model's own valve is linear with C_v 62.1 at 100 %.

5 · SIS and BPCS separation

Each SIS function should have its own sensor, logic solver and final element (IEC 61511), so that one failure cannot both cause the demand and defeat the protection. Computed from this list: system of each sensor and final element, and the DCS tags that read the same sensor.

Function	Sensor (system)	Also read by the DCS	Final elements (system)	Verdict
LSLL-202	LS-202 (SIS)	—	XV-209 (SIS), XV-202 (SIS)	✓ own SIS sensor
LSHH-203	LS-203 (SIS)	—	XV-201 (SIS), XV-209 (SIS), XV-202 (SIS)	✓ own SIS sensor
HS-ESD	HS-ESD (SIS)	—	XV-201 (SIS), XV-209 (SIS), XV-202 (SIS)	✓ own SIS sensor
FSLL-201	FT-201 (DCS)	FIC-201	HY-201B (DCS)	✗ sensor FT-201 is a DCS instrument (shared with FIC-201); final element on the DCS
PDSL-230	PDT-230 (SIS)	PDI-230	HY-202B (DCS)	✗ own SIS sensor, repeated to the DCS (PDI-230): acceptable one way; final element on the DCS
PSHH-221 (rec.)	PT-221 (SIS)	—	XV-209 (SIS), XV-202 (SIS)	✓ own SIS sensor

The shut-off valves XV-201, XV-202 and XV-209 are in series with FV-201, TV-202 and TV-208 (separate final elements on the same line). The column pressure has two transmitters (PT-210, PT-211) but no SIS function; the recommended PSHH-221 on PT-221 closes this gap.

6 · Checks against the model and the other documents

19 ✓, 4 ✗. A ✗ is a finding of this list and is carried as an engineering query (section 9). The same checks are recomputed live in the Instrumentation & control tab of the simulator page.

Check	Instrument list	Model / other document	Result	Note
INL alarm limits = simulator alarm list ALM	15 of 15 tags equal	15 alarms	✓	
Trip setting LSLL-202 = K.Lsll	< 5 % for 5 s	5 %	✓	
Trip setting LS-202 = K.Lsll	< 5 %	5 %	✓	
Trip setting LSHH-203 = K.Lshh	> 92 % for 5 s	92 %	✓	
Trip setting LS-203 = K.Lshh	> 92 %	92 %	✓	
Trip setting FSLL-201 = K.fsll	< 15 m³/h for 5 s	15 m³/h	✓	
PSV-201 set pressure = K.Psv	1.80 MPa g	1.80 MPa g	✓	
Ranges cover the normal value at design	all inside	design snapshot	✓	
Ranges cover set points, alarm and trip limits	all inside	INL	✓	
Flow meters: normal flow 20–90 % of range	FT-217 9 %, FI-217 9 %	orifice turndown	✗	query EI-06
Every P&ID bubble on the list	all 32 tags	BUB	✓	
Every DCS indicator on the list	all 27 tags	IND (MARGIN is a calculation)	✓	
Every controller of the model on the list	9 controllers	sim.C	✓	
Tag numbers unique	113 unique		✓	
Model liquid valves (not choked) behave as fixed-Cv linear valves	FV-201 ±0.0 %, FV-204 ±0.9 %	Cv ÷ opening over the 3 cases	✓	
Selected valves inside the 10/20–90 % window in all cases	PV-210 8/57/2 %	7 valves	✗	query EI-05
No flashing or cavitating liquid valves	LV-202, LV-203	IEC 60534 regime	✗	queries EI-03, EI-04
SIS functions on their own sensors and final elements	FSLL-201, PDSL-230	5 functions	✗	query EI-01
Alarm before trip: LIC-202 LAL above LSLL-202	25.0	5.00	✓	
Alarm before trip: LIC-203 LAH below LSHH-203	75.0	92.0	✓	
Alarm before trip: FIC-201 FAL above FSLL-201	45.0	15.0	✓	
Alarm before trip: PIC-210 PAH below PI-211 PAH	1.76	1.78	✓	
Alarm before trip: PI-211 PAH below PSV-201 set	1.78	1.80	✓	

7 · Changes made to the first draft

The first draft (epc/epc.json, 110 rows) was refined by tabs/einst/work/mk_inl.js:

Tag	Change	Reason
ST-201	moved out of the INL	manual valve / steam trap: piping special item (valve list), not an instrument
ST-202	moved out of the INL	manual valve / steam trap: piping special item (valve list), not an instrument
HV-202	moved out of the INL	manual valve / steam trap: piping special item (valve list), not an instrument
HV-209	moved out of the INL	manual valve / steam trap: piping special item (valve list), not an instrument
HV-210	moved out of the INL	manual valve / steam trap: piping special item (valve list), not an instrument
HV-216	moved out of the INL	manual valve / steam trap: piping special item (valve list), not an instrument
HY-201A	renamed from HS-201A	tag used twice in the draft (HS-201B / HS-202B are the auto-start enable switches on the P&ID)
HY-201B	renamed from HS-201B	tag used twice in the draft (HS-201B / HS-202B are the auto-start enable switches on the P&ID)
HY-202A	renamed from HS-202A	tag used twice in the draft (HS-201B / HS-202B are the auto-start enable switches on the P&ID)
HY-202B	renamed from HS-202B	tag used twice in the draft (HS-201B / HS-202B are the auto-start enable switches on the P&ID)
XA-201A	added	motor-trip status from the MCC; without it a stopped motor and a lost run signal look the same
XA-202A	added	motor-trip status from the MCC; without it a stopped motor and a lost run signal look the same
XA-202B	added	motor-trip status from the MCC; without it a stopped motor and a lost run signal look the same
LI-201	added	bubble on the P&ID (rev B) and in the alarm list (LAL/LAH-201) but not in the draft
LS-202	I/O DI → AI	the SIS reads a level in % (LSLL/LSHH bubbles show it); an analog transmitter gives diagnostics and a comparison with LT-20x
LS-203	I/O DI → AI	the SIS reads a level in % (LSLL/LSHH bubbles show it); an analog transmitter gives diagnostics and a comparison with LT-20x
FE-202 … FE-217	location filled in	P&ID line taken from the indicator of the same loop
HS-RST	added	the SIS reset command of the model (bmsCmd reset) needs a hard-wired input
PT-221	added as recommended	Safety layers tab recommendation; counted separately
PSHH-221	added as recommended	Safety layers tab recommendation; counted separately
XV-222	added as recommended	Safety layers tab recommendation; counted separately
ZSC-222	added as recommended	Safety layers tab recommendation; counted separately
PSV-201	orifice note	the draft said "API L orifice from the Design tab"; the relief load basis needs 4M6 (ER-01) — kept as L until the P&ID is revised

8 · Assumptions

- One field transmitter behind every DCS indicator (P&ID note 13); orifice flow elements for liquids, steam and cooling water; RTDs in thermowells; DP level transmitters.
- SIS logic solver separate from the DCS, own UPS, de-energise-to-trip; trips in the logic solver on analog transmitters LS-202 / LS-203 (drawn as switches).
- Installed spare I/O 20 %; cards of 16 (DCS) and 8 (SIS) channels.
- Globe control valves, F_L 0.90, x_T 0.72, inherent rangeability 50:1, valve = line size (F_p = 1); γ 1.30 for LP steam, 1.20 for the vent gas with Z 0.90; P_c 4 248 / 3 796 kPa a for the light / heavy pseudo-components; molar masses and densities as DC3-V1-PE-HMB-001.
- Standard rated C_v series (R10 rounded) and the characteristic rule as stated in section 4.
- Recommended additions (PT-221 / PSHH-221 at 1.77 MPa g for 3 s, XV-222 / ZSC-222) as proposed in the Safety layers tab; not in the base scope.

9 · Engineering queries raised

Open until answered by the discipline named; the Project & deliverables tab collects them for the whole unit.

No.	Subject	Query	Proposed answer	Status
EI-01	Auto-start functions share BPCS transmitters	FSLL-201 (auto-start of P-201B) uses FT-201, the transmitter of FIC-201, and PDSL-230 shares PDT-230 with the DCS indicator PDI-230. The draft list places both functions in the SIS while the pump start outputs (HY-201B, HY-202B) and FT-201 are on the DCS.	Auto-starts of spares are not SIFs and are not credited as independent layers: implement FSLL-201 and PDSL-230 in the DCS with the existing transmitters (re-allocate 2 functions, PDT-230 to the DCS), or give each its own transmitter if they are ever credited in LOPA.	OPEN
EI-02	LS-202 / LS-203 drawn as switches	The trip sensors are drawn as level switches but the SIS and the screens use a level in % (LSLL-202 at 5 %, LSHH-203 at 92 %).	Specify 4–20 mA level transmitters for the SIS (tags LT-202S / LT-203S, or keep LS with a note) with deviation alarms against LT-202 / LT-203.	OPEN
EI-03	LV-202 flashing service	The bottoms leave the sump at their bubble point (109.0 °C, Pv 1819.2 kPa a) and LV-202 drops them to the debutanizer at 0.48 MPa g: the valve is choked and flashing in every case (required Cv 11.42 at design). The model's square-root valve law does not choke.	Angle-body valve with hardened trim, outlet expanded to the two-phase line size, located close to the debutanizer inlet; size the downstream line for two-phase flow (Process).	OPEN
EI-04	LV-203: propane flashes into storage	The propane product leaves V-202 at 39.4 °C with a bubble pressure of 1621.8 kPa a, but the model's storage back-pressure is 0.90 MPa g (1001.3 kPa a): LV-203 is choked and the rundown flashes.	Confirm the storage pressure (or add a product cooler) so the rundown stays liquid; until then size LV-203 for flashing service.	OPEN
EI-05	PV-210 rangeability	The vent flow varies from 21.4 to 727.9 kg/h between turndown and maximum feed (Cv ratio 22.8:1, choked in every case). A single linear valve of Cv 4 would be 2.5 % open at turndown and 57.1 % at maximum feed.	Use a split-range pair (small + large PV) or an equal-percentage valve with a high rangeability trim; the maximum-feed case also needs the condenser margin (EP-07).	OPEN
EI-06	FT-217 vent flow meter range	FI-217 is ranged 0–800 Nm³/h but reads 73 Nm³/h at design (9 % of span), 29 at turndown and 438 at maximum feed: an orifice meter (about 4:1 usable turndown) cannot cover this.	Use a high-turndown meter for the vent gas (thermal mass or ultrasonic, 0–500 Nm³/h) or two ranges; the vent flow is the operator's main indication of non-condensables and of condenser blanketing.	OPEN
EI-07	XV limit switches	XV-201, XV-202 and XV-209 have a closed limit switch (ZSC) only.	Add open limit switches (ZSO) so the SIS can confirm the valves are open before start-up and record partial-stroke tests.	OPEN